INCH-POUND

MIL-PRF-1/1350E
8 July 1999
SUPERSEDING
MIL-E-1/1350D
20 February 1976

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 8295A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Pentode, external anode, ceramic-metal.

See figure 1.

Mounting position: Any.

Weight: 3 pounds (1.36 kg) nominal.

ABSOLUTE RATINGS:

Parameter:	F	Ef	Eb	Ec1	Ec2	Ec3	Ib
Unit:	MHz	V ac	V dc	V dc	V dc	V dc	mA dc
Class AB1 (rf or audio)							
Maximum:	30	6.3	3,000	---	600	100	800
Minimum:	---	5.7	---	---	---	---	---
Test conditions:	---	6.0	2,500	Adj	500	0	400

ABSOLUTE RATINGS:

Parameter:	Ic1	Pp	Pg1	Pg2	tk	Anode core and seal temp	Cooling
Unit:	mA dc	W	W	W	sec	°C	1/
Class AB1 (rf or audio)							
Maximum:	0	1,000	0	30	---	250	---
Minimum:	---	---	---	---	180	---	---
Test conditions:	---	---	---	---	300	---	2/

GENERAL:

Qualification: Required.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
Conformance inspection, part 1									
Pulsing emission (sinusoid)	1231	---	prr = 10 (min); eb = ec2 = ec1 = 1,000 v	0.65	II	is	70	---	a
Electrode voltage (grid)	1261	---		0.65	II	Ec1	-85.0	-115.0	V dc
Total grid current	1266	5/		0.65	II	Ic1	---	-50	µA dc
Electrode current (screen)	1256	---		0.65	II	Ic2	---	15	mA dc
Conformance inspection, part 2									
Heater current	1301	---		---	---	If	7.7	8.7	A ac
Direct-interelectrode capacitance (ground cathode connection)	1331	---		---	---	Cgp Cin Cout	--- 36.0 16.5	0.12 44.0 20.5	pF pF pF
Amplification factor	1316	---	g1 to g2 amplification; Ec1/Ic2 = 60 mA dc; Eb = 0	---	---	Mu	3.0	3.8	---
Low-frequency vibration	1031	---	No voltages applied	---	---	---	---	---	---
Low-frequency vibration end-points:	---								
Electrode voltage (grid)	1261	---		---	---	Ec1	-85.0	-115.0	V dc
Total grid current	1266	---		---	---	Ic1	---	-50	µA dc
Current division (short pulse, method B)	1372	---	Eb = Ec2 = 500 V dc Ec1 = -200 V dc; egk/ib = 2.5 a	---	---	egk ic2	--- ---	0 400	v ma

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
Conformance inspection, part 3									
Stability	---	3/	Eb = 3,000 V dc; Ec1/Ibo = 220 mA dc at t = 0; Read ΔIb at t = 180 sec	---	---	ΔIb t	---	22	mA dc
Linear amplifier power output and distortion	2204	3/	Class AB1 amplifier; Eb = 3,000 V dc; Ec1/Ibo = 220 mA dc; Eg1/Ib = 800 mA dc (1-tone); Rℓ = 2,000 ± 100 ohms; Rg = 1,000 ohms (max); anode tank Q = 10 to 15	---	---	Po 3rd IM 5th IM	1.350 -20 -30	--- --- ---	W dB dB
Life test	---	4/	Group C; linear amplifier Power output and distortion except, Eg1/Po = 1,400 W (min) (1-tone); t = 500 hours	---	---	---	---	---	---
Life-test end points:	2204	---							
Linear amplifier power output and distortion				---	---	Po 3rd IM 5th IM	1,250 -19 -29	--- --- ---	W dB dB

1/ With an anode dissipation of 1,000 watts and an incoming air temperature of 50°C at sea level, a minimum of 24 cfm of air shall be passed through the anode cooler. In all applications, an air-system socket, such as the EIMAC SK-184 or SK-184A, or equal, should be used. With an airflow of 24 cfm, the static pressure drop for the tube and socket is approximately 0.13 inch of water. Cooling air should be applied before or simultaneously with the application of electrode voltages, including the heater, and may be removed simultaneously with them. In cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

2/ During all electrical tests involving application of heater voltage, the use of an air-system socket is permissible, with forced-air cooling of 24 cfm maximum for air at 25°C at sea level. Correction may be made for higher altitude or temperature.

3/ The following tests, listed under conformance inspection, part 3, shall be performed every 3 months, with sample size and acceptance criteria as follows:

n1 = 4	c1 = 0	where c2 represents the total failures for the
n2 = 4	c2 = 1;	first and second samples combined.

In case of a failure on a stability-test sample, the lot shall be reprocessed and a new sample submitted. the linear amplifier power output and distortion test shall be performed on the sample that is used for the stability test. In case of a failure of a linear amplifier power output and distortion test sample, this test shall become a conformance inspection, part 2, with an acceptance level of 6.5, inspection level of S3. After three consecutive successful submissions, the test may revert to the conformance inspection, part 3 test.

4/ This is a destructive test. All voltage and current meters shall be calibrated for ± 2 percent or better accuracy as installed in the equipment. Prior to the performance of the life test, the tube shall have met the requirements of the linear amplifier power output and distortion test.

5/ This test is to be the first test performed at the conclusion of the holding period.

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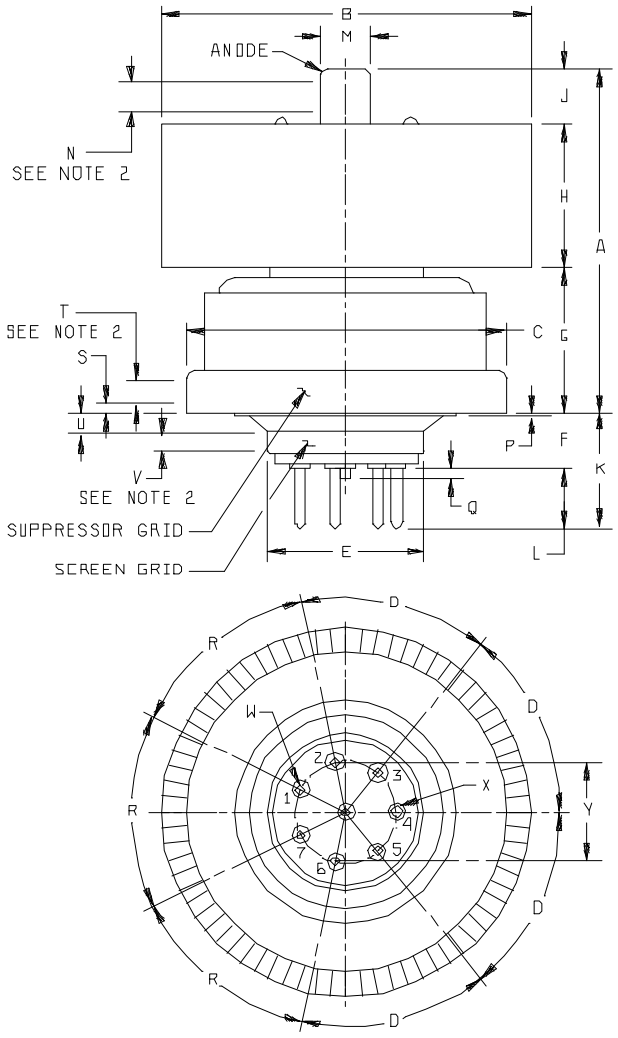

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	3.458	3.832	87.83	97.33
C	3.485	3.515	88.52	89.28
E	1.615	1.630	41.02	41.40
F	.655	.719	16.64	18.26
K	1.056	1.219	26.82	30.96
M	.559	.573	14.20	14.55
W	.056	.062	1.42	1.57
X	.120	.127	3.05	3.23
Conformance inspection, part 3 (see note 1)				
B	3.968	4.032	100.79	102.41
G	1.395	1.645	35.43	41.78
H	1.468	1.532	37.29	38.91
J	.593	.657	15.06	16.69
L	.438	.562	11.13	14.27
N	.400	---	10.16	--
P	---	.125	---	3.18
T	.250	---	6.35	---
V	.220	---	5.59	---
Reference dimensions				
D	51°		51°	
Q	.125		3.18	
R	52°		52°	
S	.125		3.18	
U	.250		6.35	
Y	1.000		25.40	

PIN CONNECTIONS	
Pin No.	Element
1	k
2	g1
3	h
4	k
5	h
6	g1
7	k
Center pin	int con
Lower ring	g2
Upper ring	g3
Cap	a

NOTES:

1. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number number of one. In the event of failure, the test will be made as a part of conformance inspection, part 2, acceptance level of 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.
2. Contact surface shall be confined to this area.

FIGURE 1. Outline drawing of electron tube type 8295A.

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Custodian:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - CR4
Navy - AS, CG, MC, OS
Air Force - 17, 99

Preparing activity:
DLA - CC

(Project 5960-3546-12)

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